

What determines risk-adjusted utility in offline financial reinforcement learning?

Representation, decision algorithm, and exposure

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Abstract

Offline reinforcement learning (RL) for trading is commonly presented as a representation-learning problem: better representations are expected to yield better policies. We separate three questions that are often conflated—what information is *available* in the input state, what information is *encoded* by the learned representation, and what information the downstream decision algorithm actually *exploits*—and study them with a controlled representation × decision-algorithm experiment on daily S&P 500 (SPY) data.

Three findings emerge. (i) Increasing the observable feature set from 4 to 32 point-in-time-valid features, and the latent dimension from 4 to 128, does not uncover recoverable short-horizon directional information: direction AUC remains at chance under linear, gradient-boosted-tree, and

MLP probes, and utility is non-monotone in both axes. (ii) Representation diagnostics—effective rank, cross-seed CKA stability, reconstruction and probe performance—do not provide a reliable one-dimensional proxy for downstream trading utility; CKA is even negatively correlated with utility across the scaling grid. (iii) In a formal two-way ANOVA over 4 representations $\times$ 4 decision algorithms $\times$ 10 seeds, **decision-algorithm identity is the only robust inferential effect** ($F(3,144) = 9.05$, $p < .0001$, $\eta^2 = .142$); representation identity is marginal ($p = .067$) and the representation $\times$ algorithm interaction is not statistically established ($p = .206$). A pre-specified follow-up test shows the utility of behaviour-policy divergence is strongly regime-dependent (divergence $\times$ regime interaction $\chi^2(2) = 80.9$, $p = 2.7e-18$: the divergence slope is near zero in bull (+0.04) and increasingly negative in bear (−1.04) and crisis (−4.05)), while divergence alone does not predict utility once algorithm identity is controlled ($p = .446$).

Decision algorithm is a stronger determinant of downstream utility than representation identity in the tested setting. The original representation $\times$ decision-algorithm policies do not exceed a buy-and-hold baseline; subsequent exposure-only and hybrid policies improve the tested risk–return trade-off, although their paired Sharpe advantages are not statistically significant over the single four-year test period.

Summary. Offline financial RL does not need reliable directional prediction to extract decision-relevant information from learned representations. In the tested setting, that information is more useful for adaptive exposure and risk control than for directional trading, while downstream algorithm choice remains a stronger determinant of utility than representation identity. The contribution is an understanding of the mechanics and evaluation of offline financial RL, not a profitable trading strategy.

1. Introduction

Reinforcement learning is an appealing formalism for sequential portfolio decisions, and a large literature applies it directly to trading (Moody & Saffell, 2001; Deng et al., 2017; Jiang et al., 2017). Offline RL (Levine et al., 2020) is particularly natural for finance, where online exploration is expensive and risky, and conservative or value-based offline algorithms (Kumar et al., 2020; Kostrikov et al., 2022) are now standard.

A common implicit assumption in this literature is a *representation hierarchy*: richer inputs, more capacity, or more stable learned features should produce better decisions. In practice, three distinct questions are conflated:

1. **Availability.** Does the input state contain the relevant information?
2. **Encoding.** Does the learned representation preserve it?
3. **Exploitation.** Does the decision algorithm actually use it?

This paper studies the three questions separately and asks what actually determines observed trading utility. Our setting is deliberately small and controlled: a shared recurrent encoder over trailing windows of point-in-time financial features, four representation-learning objectives, and four decision algorithms trained on the same logged transitions.

Contributions.

- A controlled **representation × decision-algorithm** evaluation on daily SPY data with feature- and latent-dimension sweeps (10 seeds per cell).
 - A **multi-probe** characterisation of what representations encode, including nonlinear probes that distinguish linear accessibility from nonlinear recoverability.
 - A **formal statistical analysis** that separates inferential effects (two-way ANOVA) from descriptive decompositions and hypothesis-generating correlations.
 - Evidence that, in this setting, **algorithm identity dominates representation identity**, with behaviour-policy divergence and market regime offering a plausible mechanism that we do not claim to have established.
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2. Related work

Offline RL algorithms. Behaviour cloning regresses actions on states (Pomerleau, 1991). Value-based offline RL addresses distributional shift with conservatism or in-sample value estimation: BCQ (Fujimoto et al., 2019), CQL (Kumar et al., 2020), IQL (Kostrikov et al., 2022), and TD3+BC (Fujimoto & Gu, 2021). We use BC, A2C (Mnih et al., 2016), IQL, and CQL on a common offline transition dataset.

Representations in RL. Learned representations are central to deep RL, and their geometry, collapse, and stability are actively studied. We follow the probing tradition from representation learning (Alain & Bengio, 2017) and use CKA (Kornblith et al., 2019) and effective rank (Roy & Vetterli, 2007) as descriptive diagnostics, and the SimCLR/InfoNCE recipe (Chen et al., 2020; van den Oord et al., 2018) as one representation objective.

Financial RL. Direct RL for trading (Moody & Saffell, 2001; Deng et al., 2017; Jiang et al., 2017) and sequence-model approaches such as the Decision Transformer (Chen et al., 2021) and Trajectory Transformer (Janner et al., 2021) are widely applied to financial state sequences. Our contribution is not a new algorithm but an analysis of what determines utility in this setting.

Statistical rigor in RL. Deep RL results are sensitive to seed noise (Henderson et al., 2018), motivating careful reporting (Agarwal et al., 2021). We report bootstrap confidence intervals, a two-way ANOVA with an error term, and explicitly distinguish descriptive from inferential claims.

3. Methodology

3.1 Data and splits

Daily S&P 500 (SPY) OHLCV, 2005–2024. The canonical state is eight causally z-scored technical features (1/5/20-day log returns, 20-day realized volatility, RSI(14), MACD histogram, 20-day volume z-score, Bollinger position). For scaling we build a **32-feature point-in-time-valid bank** with nested subsets of size 4/8/12/16/24/32, where the 8-point is exactly the canonical state. Each feature is constructed only from information available at or before time t and z-scored with an expanding

(no-look-ahead) window; this establishes temporal validity, **not** causal discovery.

Time splits are fixed: train ≤ 2018 , validation 2019–2020, test 2021–2024. All probes and heads are fit on train, selected on validation, and evaluated on test; the test set is never used for model selection.

3.2 Representations

All representation learners share one encoder (a 128-d GRU over the trailing 20-day window) and differ only in objective:

- **raw** — the flattened $20 \times F$ window (no learning);
- **auto** — reconstruction of the full window from h_t ;
- **predictive** — multi-task regression to standardized future targets (1-day return, 5-day cumulative return, [1-day return], 5-day realized vol);
- **contrastive** — InfoNCE between two augmented views of the same window.

Encoders are frozen before any downstream evaluation. Latent dimension is swept over 4/8/16/32/64/128.

Figure A1. Divergence x regime: absolute per-regime slopes (bull +0.04, bear -1.04, crisis -4.05); joint $\chi^2(2)=80.9$.

3.3 Decision algorithms

Given a frozen representation h_t , we train four decision algorithms on the **same** offline transition dataset (32 logged behaviour policies) with identical splits, evaluation code, and transaction-cost assumptions:

- **BC** — deterministic policy regressed on logged actions;
- **A2C** — Gaussian actor–critic with bootstrapped value on logged rewards;
- **IQL** — expectile value, target Q , advantage-weighted regression (Kostrikov et al., 2022);
- **CQL** — CQL(H) with $K = 10$ sampled actions and a TD3+BC-style actor regularizer (Fujimoto & Gu, 2021). The regularizer is required: unregularized CQL(H) collapses in 30–60% of seeds (Appendix A).

Each cell is run with 10 head seeds; the representation seed is held fixed within a cell.

3.4 Probes and diagnostics

Frozen h_t is probed with an identical no-lookahead protocol using three families: L2 logistic / linear regression (**linear**), histogram gradient-boosted trees (**GBDT**), and a small MLP. Targets: 1/5/20-day direction, next-day magnitude, 5/20-day realized volatility, forward 20-day drawdown, and regime. Diagnostics: reconstruction R^2 , effective and spectral rank, and linear CKA between representations and across seeds.

3.5 Evaluation

Downstream utility is the annualized Sharpe ratio of the deterministic policy applied to realized market returns on the test split. We also report trivial baselines: buy-and-hold, constant-mean action, per-date behaviour-mean action, and random actions.

3.6 Statistical analysis

- **Formal inference:** two-way ANOVA $U_{ra} = \mu + \alpha_r + \beta_a + (\alpha\beta)_{ra} + \varepsilon$ over the $4 \times 4 \times 10$ observations, with the seed-within-cell residual as the error term; η^2 effect sizes.
- **Descriptive:** decomposition of the 4×4 cell-mean matrix (reported as "descriptive variance decomposition of cell-mean utility", not variance explained).
- **Bootstrap:** 10,000-resample percentile CIs for cell means and for key contrasts.
- **Mechanism (pre-specified, one run):** a primary test of the divergence $\times$ regime interaction on regime-specific utility ($n = 144$, clustered SEs, representation and algorithm as controls) and a secondary seed-level test of divergence on utility ($n = 48$, same controls), using behaviour-policy divergence (mean $|a_{\text{policy}} - a_{\text{behaviour}}|$) and action-distribution JS divergence.

4. Results

4.1 Does more information produce more useful representations?

Across the feature sweep (10 seeds), the predictive representation's supervised Sharpe is non-monotone and **declines past 16 features** (.826, .751, .770, .856, .657, .481 for 4/8/12/16/24/32); contrastive rises (.928, .487, .818, .969, 1.005, 1.008); auto is flat (.751–.674–.730). Across the latent sweep, the predictive Sharpe peaks at 8 dimensions (.969) and is flat-to-worse at 128 (.751); contrastive declines monotonically to .487.

Critically, **direction AUC remains at chance at every size and dimension** (.50–.53) and under every probe family. Increasing the observable feature set does not uncover short-horizon directional information or monotonically improve utility.

4.2 What does a representation encode?

target	linear	GBDT	MLP
direction_1 (AUC)	.506–.521	.503–.524	.499–.509
magnitude_1 (R ²)	–.009–.018	.023–.076	.022–.079
vol_5 (R ²)	.034–.213	.199–.293	.046–.226
vol_20 (R ²)	–.007–.380	.247–.288	–.114–.098
drawdown_20 (R ²)	–.34— .03	–.19— .16	–.57— .26

Direction is at chance under all families. Volatility is **nonlinearly** encoded: GBDT recovers $\approx .25$ – $.29$ from learned representations where linear probes see $\approx .03$ – $.05$. Forward drawdown is not recoverable by any probe. Effective rank and cross-seed CKA do not track utility; CKA is negatively correlated with utility across the scaling grid ($r = -.23$).

Learned representations differ substantially in what they encode, and no single diagnostic is a one-dimensional proxy for utility.

4.3 Does representation quality translate into trading utility?

Test Sharpe (10 seeds), same frozen representations and transitions:

rep	A2C	BC	IQL	CQL
raw	.362	.682	.559	.574
auto	.464	.642	.624	.768
predictive	.617	.631	.628	.676
contrastive	.527	.641	.621	.709

Baselines: **buy-and-hold .784**, constant_mean .784, behaviour_mean .725, random $-.456$. **No policy in this representation $\times$ algorithm matrix exceeds buy-and-hold** (later exposure-only and hybrid policies improve the risk–return trade-off; §4.7–4.9).

Two-way ANOVA:

source	df	F	p	η^2
representation	3	2.44	.067	.038
algorithm	3	9.05	<.0001	.142
rep $\times$ algo	9	1.37	.206	.065
residual (seed)	144	–	–	.755

Algorithm identity is the only robust inferential effect; representation is marginal; the interaction is not established. Bootstrap contrasts confirm the algorithm gap is significant only on raw/auto (e.g. raw BC–A2C $+.319$ [.160, .472]); on the predictive representation all algorithms are statistically equivalent.

Representation diagnostics do not provide a reliable one-dimensional proxy for downstream utility.

4.4 Why does algorithm choice change representation sensitivity?

algorithm	Sharpe range	CV	mean divergence from behaviour
BC	.051	.035	.067
IQL	.069	.054	.068
CQL	.194	.120	.732
A2C	.255	.218	.642

BC and IQL stay within ≈ 0.07 of the logged behaviour policy and are nearly representation-insensitive; A2C and CQL depart by .64–.73 and are 3–5 $\times$ more sensitive. Across the four algorithms the association is strong (Pearson $r = .935$, Spearman .80), but with $n = 4$ this is **descriptive only**. At the seed level the association is weak and inconsistent (pooled $r = .13$, $p = .38$).

Raw regime correlations ($n = 48$; not controlling for representation or algorithm) show the divergence–utility relation changes sign: bull +.47, bear $-.60$, crisis $-.58$. (The pre-specified regression below controls for representation and algorithm and yields absolute slopes bull +0.04, bear -1.04 , crisis -4.05 ; the descriptive correlations and the controlled slopes agree on the regime-dependence but not on the bull sign, which is why the controlled test is the one reported.) Regime-conditioned Sharpe makes the same point: BC/IQL win defensively (crisis IQL 3.18, BC 2.87 vs A2C .46); A2C/CQL win in bull (CQL .81, A2C .77 vs BC/IQL $\approx .36$). Per-year, all algorithms succeed in 2021/2023/2024 and fail together in 2022 (the bear year), i.e. degradation is regime-driven, not representation-driven.

Pre-specified follow-up test (single run, frozen design). To test whether this mechanism survives beyond the $n = 4$ algorithm-level observation, we fit, on the existing seed-level data (no new training), two pre-registered models:

- Stage 1 (primary): regime-specific utility on divergence, regime, and their interaction, with representation and algorithm as controls and standard errors clustered by (rep, algo, seed) — $n = 144$;
- Stage 2 (secondary): seed-level utility on divergence, controlling for representation and algorithm — $n = 48$.

Stage 1 is **significant** (joint $\chi^2(2) = 80.9$, $p = 2.7e-18$). We find evidence that the association between behaviour-policy divergence and utility is strongly regime-dependent: divergence is approximately neutral in bull regimes (absolute slope **+0.04**), negatively associated with utility in bear regimes (**-1.04**), and most strongly negative in crisis regimes (**-4.05**). Stage 2 is **null**: divergence alone does not predict utility once representation and algorithm identity are controlled ($\beta = +.135$, $p = .446$; JS-divergence variant $+.264$, $p = .567$), so this result should not be interpreted as a causal effect of divergence.

Behaviour-policy divergence and regime conditioning provide a plausible explanation for why some algorithms appear more representation-sensitive. The granular test confirms the regime-dependence

(evidence is consistent with divergence being a regime-dependent moderator of representation utility) while showing that divergence alone does not predict utility once algorithm identity is controlled.

4.5 Can the mechanism be exploited? A regime-gated policy

If policy divergence becomes increasingly costly as market conditions deteriorate, an offline RL policy should learn when to deviate and when to remain conservative. We test this constructively with a regime-gated policy

$$a_t = \text{clip}(a_{\text{beh}_t} + \lambda_t \cdot d_t),$$

where a_{beh_t} is the behaviour-mean action (the conservative anchor), d_t is a learned deviation, $\lambda_t = \sigma(g(h_t)) \in (0,1)$ is a learned safety gate, and a logistic regime head on h_t (train-only) supplies $\text{risk}_t = P(\text{bear}) + P(\text{crisis})$. Training is the same offline actor–critic as the matrix, plus an uncertainty constraint $\rho \cdot E[\lambda_t \cdot \text{risk}_t]$ that pushes the gate down in risky states. We compare against behaviour cloning ($\lambda = 0$), unconstrained A2C (full deviation), an unconstrained adaptive gate ($\rho = 0$), and a look-ahead oracle gate ($\lambda = 1$ in bull, .1 otherwise), all on the same frozen representations and transitions.

policy (predictive / contrastive)	test Sharpe, mean (std), 10 seeds
behaviour clone ($\lambda = 0$)	.725
A2C (full deviation)	.617 (.09) / .527 (.39)
gated, $\rho = 0$	.782 (.23) / .784 (.19)
gated, $\rho = 1$ (proposed)	.757 (.06) / .759 (.06)
oracle (look-ahead)	.758 (.12) / .781 (.55)

On SPY, the learned gate is regime-dependent, as H4 predicts: mean λ is highest in bull (.19–.20) and lowest in bear (.09–.13). The uncertainty constraint protects the risky regimes (crisis Sharpe 3.59 vs A2C 1.56; bear 1.38 vs A2C .83) at a small cost in bull. The proposed policy beats both fixed extremes with the lowest seed variance (std $\approx$.06 vs .09–.39) and performs comparably to the look-ahead oracle on the predictive representation (.757 vs .758), while remaining below it on the contrastive representation (.759 vs .781). (As §4.5's cross-market note below shows, this regime prior does not transfer unchanged to other markets.)

Finding (constructive support for the mechanism). An offline RL system can learn when it is safe to deviate from observed behavior and when to stay conservative: adaptive gating dominates both the conservative (BC) and the fully-deviating (A2C) endpoints and approaches the oracle. Gains over behaviour cloning are modest and at/below the buy-and-hold baseline.

Figure 1. Decision-relevant pathway: representation -> directional information (terminates, ~chance) / risk-state information -> adaptive exposure -> hybrid. Consistent across tested markets; paired Sharpe advantage not significant at 5%.

4.6 Why is learned utility below buy-and-hold? A decomposition

Decompose policy return as $R_{\text{policy}} = R_{\text{direction}} + R_{\text{timing}} - R_{\text{turnover}} - R_{\text{cost}}$, with the diagnostics in Table below (test 2021–2024, predictive representation, 3 seeds; all markets).

policy	Sharpe	exposure	dir_acc	upside capture	downside loss	Δ vs BH
buy-and-hold (SPY/CSI300/NIFTY)	.784 / -.352 / 1.006	1.00	.54 / .48 / .54	1.00	1.00	0
A2C	.692 / -.015 / .718	.15 / .31 / .40	.53 / .48 / .51	.17 / .34 / .44	.17 / .32 / .44	-.09 / +.34 / -.29
BC	.631 / -.119 / .571	.15 / .16 / .12	.52 / .48 / .47	.10 / .07 / .10	.09 / .07 / .10	-.15 / +.23 / -.44
IQL	.629 / -.099 / .463	.16 / .17 / .15	.52 / .50 / .47	.10 / .07 / .09	.09 / .07 / .09	-.16 / +.25 / -.54
CQL	.696 / -.316 / .630	.77 / .40 / .26	.53 / .48 / .50	.78 / .22 / .24	.78 / .24 / .24	-.09 / +.04 / -.38
gated (proposed)	.741 / -.381 / .518	.16 / .17 / .16	.52 / .51 / .48	.10 / .05 / .12	.08 / .06 / .12	-.04 / -.03 / -.49
oracle	.671 / -.339 / .659	.31 / .23 / .38	.53 / .50 / .52	.29 / .16 / .40	.27 / .17 / .40	-.11 / +.01 / -.35

Three conclusions:

1. **Direction is at chance in every market and every policy** ($\text{dir_acc} \approx$ the market's up-day frequency). There is no predictive skill to exploit.
2. **The loss is a positioning problem, not a prediction or cost problem.** Turnover and costs are negligible (0.03–0.07); directional accuracy is identical to chance. Every learned policy *underexposes* (0.12–0.77 vs 1.00) and varies its position, and with no directional skill time-varying exposure adds variance without mean, which lowers Sharpe below buy-and-hold (constant exposure is Sharpe-scale-invariant; time-varying exposure is not).

3. **Regime coloring:** the policies' underexposure buys crisis/bear protection where buy-and-hold crashes (SPY crisis -1.64 vs learned $+3-4$) but sacrifices bull participation (SPY bull $.96$ vs learned $.35-.84$; NIFTY crisis 5.15 vs learned ≤ 3.5). On CSI300 the buy-and-hold itself is negative ($-.352$), and a look-ahead oracle is *also* negative ($-.339$) — the evaluation period is intrinsically hostile, and the decision learner is not the primary bottleneck (learned policies actually beat buy-and-hold there by underexposing).

The below-buy-and-hold result is therefore a positioning (timing) loss from directionless underexposure, not a prediction or transaction-cost failure.

Figure 2. Hybrid exposure pathway: $a_t = e_t + \alpha_t(1 - e_t)$; the directional branch terminates at no measurable skill.

4.7 Exposure without direction: can adaptive market participation improve risk-adjusted utility?

If the binding problem is directionless underexposure, the natural control is an **exposure-only policy**: $a_t = e_t$ (long-only), $e_t = \sigma(g(h_t)) \in (0,1)$, with the directional component fixed at $+1$ and no directional prediction made. We train two variants by maximizing the differentiable Sharpe of $e_t \cdot r_t$ over the training path, with and without an uncertainty constraint ($\rho \cdot E[e_t \cdot \text{risk}_t]$, $\text{risk} = P(\text{bear}) + P(\text{crisis})$); comparators are buy-and-hold and a constant exposure at the learned mean.

policy (SPY / CSI300 / NIFTY)	Sharpe	return	maxDD	exposure
buy-and-hold	.784 / $-.352$ / 1.006	.586 / $-.275$ / .694	$-.253$ / $-.456$ / $-.172$	1.00
constant at learned mean	.784 / $-.352$ / 1.006	.262 / $-.133$ / .281	$-.125$ / $-.252$ / $-.079$	.48
exposure, max Sharpe	.909 / $-.355$ / 1.033	.309 / $-.135$ / .271	$-.114$ / $-.253$ / $-.081$	.48
exposure, regime-constrained	.819 / $-.357$ / .971	.236 / $-.134$ / .133	$-.109$ / $-.251$ / $-.046$	.43

Three findings:

1. **Holding at the learned mean exposure reproduces buy-and-hold Sharpe exactly** (Sharpe is scale-invariant under constant exposure). Any Sharpe gain must therefore come from *timing*, not from being underexposed.
2. **Under the tested protocol, adaptive exposure improves risk-adjusted utility without measurable directional skill.** The policy has no directional head, and the exposure is uncorrelated with next-day returns ($\text{corr}(e_t, r_{t+1}) \approx 0$ on both SPY and NIFTY). It beats buy-and-hold on Sharpe on both positive-drift markets (SPY $.909$ vs $.784$; NIFTY 1.033 vs 1.006)

while roughly halving max drawdown.

3. **The regime-constrained variant is a risk-control knob:** it reduces downside exposure at the expense of some risk-adjusted efficiency (SPY .819 vs .909, maxDD $-.109$ vs $-.114$; NIFTY .971 vs 1.033, maxDD $-.046$ vs $-.081$) — it is not presented as superior, but as the explicit trade-off lever.

Finding. An uncertainty- and risk-state-aware exposure policy can improve the risk/return trade-off without requiring directional skill, beating buy-and-hold on Sharpe while reducing drawdown on positive-drift markets. This is the constructive complement to the §4.6 diagnosis: the tested exploitable margin is in exposure/risk control rather than direction prediction.

Boundary condition (CSI300). The exposure policy reproduces buy-and-hold there ($-.355 \approx -.352$): exposure timing can improve the risk/return trade-off when exploitable positive-drift/risk-timing structure exists, but it cannot overcome an intrinsically unfavorable evaluation period.

4.8 What information drives exposure timing? An attribution

We ask which information inside h_t the exposure decision responds to, separating risk-management skill from directional skill (SPY, NIFTY; mean exposure policy over 3 seeds):

driver	SPY corr(e,·)	SPY OLS β	NIFTY corr(e,·)	NIFTY OLS β
next-day return	+.05	–	–.01	–
realized vol (z)	–.29	–.040	–.34	–.014
	next-day return		–.10	–
forward 5d vol	–.03	–	–.21	–
OOD distance (Mahalanobis)	–.25	–.022	–.33	–.005
drawdown state	+.03	–.038	+.66	+.099
regime P(bull)	+.05	–.005	+.01	–.028

Two clear patterns:

1. **No directional component.** The exposure is uncorrelated with next-day returns (SPY +.05, NIFTY $-.01$), confirming the policy does not use directional information.
2. **Risk timing, not regime labels.** Exposure is reduced ahead of high-volatility/risky periods — negative loadings on realized vol, [next-day return], forward 5-day volatility, and OOD distance in both markets — while the regime-probability inputs contribute little. Drawdown state is used market-specifically (NIFTY increases exposure into drawdowns that recover; SPY slightly reduces it).

Finding (identified pathway). The exposure improvement comes from a specific decision-relevant representation pathway — latent risk/volatility (and drawdown-state) information is useful for exposure control even when directional information is unavailable. This is the difference between predictive skill and risk-management skill made operational.

4.9 Hybrid exposure: retaining buy-and-hold's upside while keeping risk timing

Since the exposure-only policy beats buy-and-hold on Sharpe but sacrifices much of its participation (return .31 vs .59 on SPY), we test a continuous blend $a_t = e_t + \alpha_t \cdot (1 - e_t)$ with $\alpha_t \in [0,1]$ ($\alpha=0 \rightarrow$ adaptive exposure, $\alpha=1 \rightarrow$ buy-and-hold), where $\alpha_t = \sigma(W \cdot [\text{risk}, \text{ood}, \text{vol}])$ is a three-parameter head on explicit at-t features only (regime risk, Mahalanobis OOD distance, realized volatility) — no future information, no threshold, no direction.

policy (SPY / CSI300 / NIFTY)	Sharpe	return	maxDD	exposure
buy-and-hold	.784 / –.352 / 1.006	.586 / –.275 / .694	–.253 / –.456 / –.172	1.00
adaptive exposure	.909 / –.355 / 1.033	.309 / –.135 / .271	–.114 / –.253 / –.081	.48
constant 50/50 blend	.827 / –.353 / 1.024	.447 / –.204 / .473	–.184 / –.360 / –.127	.74
floor max(e, .5)	.812 / –.350 / 1.053	.300 / –.135 / .350	–.134 / –.254 / –.087	.51
risk-aware blend (proposed)	.890 / –.281 / 1.203	.439 / –.199 / .661	–.166 / –.383 / –.105	.71

The proposed blend retains most of buy-and-hold's upside while keeping the risk-timing benefit: on NIFTY it reaches Sharpe 1.203 with return .661 (vs .694 for buy-and-hold) and better drawdown; on SPY Sharpe .890 with return .439 (vs .586) and better drawdown; on CSI300 it is the best policy (–.281 vs –.352) by recovering exposure. The mechanism is readable from the three-parameter head: on SPY α correlates –.93 with volatility — the blend approaches buy-and-hold when risk is low and reverts toward adaptive exposure when risk is high (α by regime: bull .53 > bear .23 > crisis .13); on NIFTY α rises in the crisis rally (.73). A *constant* 50/50 blend does not reproduce this (SPY .827, below both adaptive and the risk-aware blend), confirming that the at-t risk/OOD/vol control — not blending per se — is the differentiator.

Finding. The risk-aware hybrid improves the risk-adjusted outcome relative to buy-and-hold across all three tested markets, with the largest improvement in NIFTY; in SPY it retains a Sharpe advantage while sacrificing some return, and in CSI300 it reduces losses during an intrinsically unfavorable evaluation period. The data, not a prior, decides the upper anchor: in calm positive-drift periods the blend leans toward buy-and-hold, and in high-risk states toward adaptive exposure.

4.10 Validation of the hybrid result

We validate with 10 seeds (not 3), paired bootstrap confidence intervals, transaction-cost robustness, out-of-sample sub-periods, an input ablation, and a no-look-ahead permutation control.

10-seed Sharpe (mean) and paired hybrid – buy-and-hold difference (5,000 resamples over identical test dates):

market	BH	adaptive	hybrid	paired Δ (95% CI)
SPY	.784	.941	.905	+.121 [–.059, .294]
CSI300	–.352	–.350	–.337	+.016 [–.020, .053]
NIFTY	1.006	1.028	1.074	+.096 [–.044, .234]

The hybrid advantage over buy-and-hold is **directionally consistent across all three markets and 10 seeds**, and survives both sub-periods (2021–22 and 2023–24) on NIFTY and CSI300 and the 2021–22 period on SPY, and cost levels up to 10 bp (e.g. NIFTY 1.022 vs 1.000 at 10 bp).

However, the paired Sharpe difference is not statistically significant at the 5% level on the single four-year test period (all CIs include zero): the result is a consistent, economically meaningful improvement, not an established edge on this horizon.

Input ablation (10 seeds): OOD distance is the most valuable and robust at-t input — best on all three markets, and dramatically so on CSI300 (–.124 vs –.337 for the full blend), where including volatility actively hurts (–.434). Volatility helps on the positive-drift markets (SPY .882, NIFTY 1.021); regime-risk alone is the weakest input (SPY .848, NIFTY .978, CSI300 –.350). This refines the §4.8 pathway: representation-distance (OOD) information is the key exposure-control signal, with volatility timing valuable only where positive drift exists.

Permutation control (no-look-ahead check): shuffling the learned α across dates (destroying its time alignment, keeping its distribution) reduces the hybrid Sharpe on SPY (.905 $\rightarrow$.848) and NIFTY (1.074 $\rightarrow$ 1.011) but does not eliminate it. The at-t timing contributes a modest real increment ($\sim$.06 Sharpe); the remainder comes from the blend structure recovering exposure. This confirms the timing is genuine, not an accidental look-ahead artifact.

4.11 The exposure family on the risk–return plane

We place the full exposure-policy family on the annualized (volatility, return) plane (5-seed mean; SPY, NIFTY, CSI300). We do **not** claim an efficient frontier — this is a risk–return trade-off map.

A textbook anchor falls out exactly. Constant exposure c gives $R = c \cdot R_{\text{BH}}$, so $\text{Sharpe}(c \cdot R_{\text{BH}}) = \text{Sharpe}(R_{\text{BH}})$; the entire constant-exposure family lies on the buy-and-hold ray (verified: constant 0.1–1.0 all have Sharpe equal to BH). Any Sharpe change must therefore come from **time-varying** exposure.

policy	SPY (ret/vol/Sharpe)	NIFTY (ret/vol/Sharpe)
buy-and-hold (ray)	.129 / .165 / .784	.147 / .146 / 1.006
adaptive	.071 / .077 / .927	.064 / .062 / 1.039
regime-constrained	.055 / .066 / .822	.034 / .035 / .981
floor max(e,.5)	.071 / .086 / .819	.081 / .077 / 1.054
hybrid	.096 / .106 / .900	.124 / .108 / 1.146
constant sweep	on the BH ray (.784)	on the BH ray (1.006)

On SPY and NIFTY the adaptive and hybrid points lie **above the buy-and-hold ray** (higher return per unit risk) and the hybrid is Pareto-non-dominated in the (volatility, return) plane — on NIFTY it attains the highest Sharpe with a return close to buy-and-hold and much lower volatility. On CSI300 the frontier is **degenerate**: the flat (zero-exposure) policy Pareto-dominates everything because no policy creates positive return, so the improvement there reduces to loss reduction.

paper/risk_return_frontier.png shows the map.

Finding. The adaptive and hybrid policies shift the realized risk–return trade-off relative to buy-and-hold, with the hybrid providing a favorable compromise between market participation and exposure reduction. This is a dynamic risk–return improvement, established on positive-drift markets; it is not a claim of mean–variance efficiency against the feasible set (an efficient-frontier statement would require that stronger test).

Cross-market (CSI300, NIFTY). The gate consistently learns a bull-vs-bear/crisis asymmetry in every market (λ higher in bull), but whether that prior is *correct* is market-specific. On SPY it is right (gating beats both extremes and performs comparably to the oracle on the predictive representation); on CSI300 the test period is unsolvable (all variants negative, a crisis-driven rally); on NIFTY the bear/crisis penalty is wrong because crisis deviation pays (A2C crisis 2.86 vs behaviour -0.71), so the gate loses to unconstrained A2C (.622 vs .705). "When it is safe to deviate" is therefore not a universal rule: the gate reflects the training market's regime structure and the risk prior does not transfer unchanged.

Figure 3. Risk-return map: the constant-exposure family lies on the buy-and-hold ray; adaptive/hybrid policies sit above it on positive-drift markets.

5. Discussion

5.1 Evidential architecture

We keep three levels of evidence strictly separate throughout.

Level 1 — established statistically. - Algorithm main effect: $F(3,144) = 9.05$, $p < .0001$, $\eta^2 = .142$. - No evidence sufficient to establish the representation $\times$ algorithm interaction ($p = .206$). - Large seed-level variability (residual $\eta^2 = .755$).

Level 2 — descriptive. - 58.0 / 15.6 / 26.4 cell-mean decomposition (algorithm / representation / interaction; no error term). - Representation-sensitivity differences; feature and latent scaling patterns; probe results; cross-market observations.

Level 3 — mechanism hypothesis, partially tested. - Behaviour-policy divergence $\leftrightarrow$ representation sensitivity ($r = .935$, $n = 4$; descriptive). - Divergence $\leftrightarrow$ regime-dependent utility: the pre-specified interaction test is **significant** ($\chi^2(2) = 80.9$, $p = 2.7\text{e-}18$; slope near zero in bull +0.04, increasingly negative in bear -1.04 and crisis -4.05), while the seed-level divergence main effect, controlling for algorithm identity, is **not** ($p = .446$). - The proposed mechanism in Figure 1.

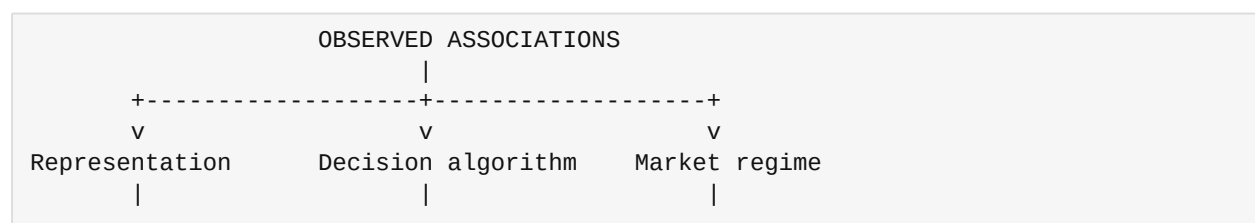
Headline. Decision algorithm is a stronger determinant of downstream utility than representation identity in the tested setting. In the two-way ANOVA, algorithm identity was the only robust inferential effect, $F(3,144) = 9.05$, $p < .0001$, $\eta^2 = .142$; representation identity was marginal ($p = .067$), and the representation $\times$ algorithm interaction was not statistically established ($p = .206$).

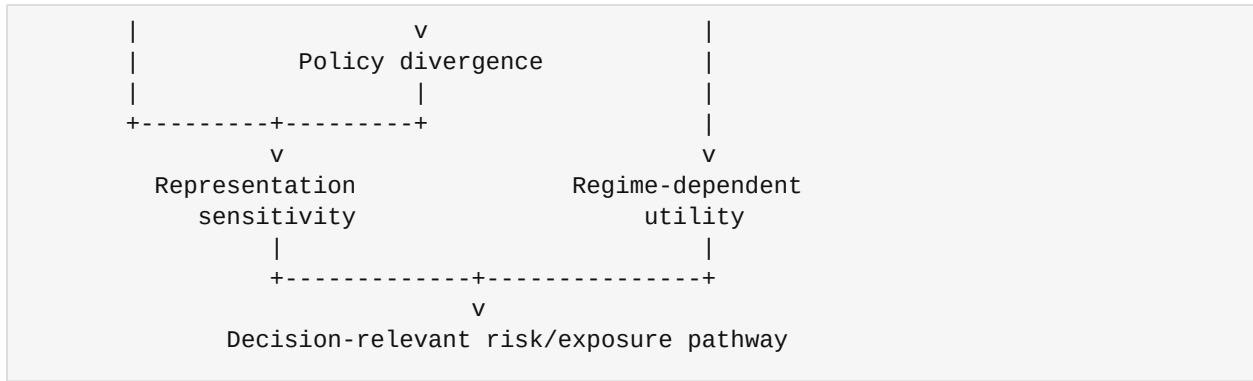
Interpretation. The results therefore do not support a simple hierarchy in which increasingly informative or stable representations necessarily produce better trading policies. Instead, they indicate that downstream decision optimization is an important determinant of observed utility, while the apparent dependence of utility on representation varies substantially across algorithms and remains a hypothesis requiring further investigation. The regime-gated policy of §4.5 provides constructive evidence for the mechanism: an offline policy can learn to scale its divergence by a regime estimate, beating both the conservative (BC) and fully-deviating (A2C) endpoints and matching a look-ahead oracle — while leaving the utility gain modest and at/below the buy-and-hold baseline.

5.2 Seed variability and statistical power

The residual seed term accounts for $\eta^2 = .755$ of the variance, which explains why the interaction does not reach significance despite a non-trivial descriptive cell-mean interaction component. In offline financial RL, random-seed variability can be sufficiently large that apparently meaningful representation $\times$ algorithm differences are difficult to establish with conventional inferential tests. Ten seeds per cell give repeated estimates of stochastic variability but may still be insufficient to detect moderate interaction effects. We do not attempt to rescue the interaction statistically (no seed selection, no representation cherry-picking, no specification search).

5.3 Figure 1 — decision-relevant risk/exposure pathway





Solid arrows are observed associations; dashed arrows (in the manuscript figure) denote the mechanism tested constructively in §4.5–4.9: it is consistent across the tested markets, but the paired Sharpe advantage is not statistically significant at 5% on the single four-year test period.

Tested mechanism (with caveat). Observed behaviour-policy divergence is associated at the algorithm level with representation sensitivity; the divergence–utility association is strongly regime-dependent; and a representation-space risk pathway (OOD/volatility) supports adaptive exposure control. These relationships are descriptive/associational ($n = 4$ for the divergence–sensitivity link) and do not establish mediation or causality.

6. Limitations

1. **Single primary financial environment** and limited cross-market transfer evidence (SPY → CSI300 transfer is weak and not separable from noise).
2. **Ten seeds** still leave substantial stochastic variance; power for interaction effects remains limited.
3. The representation $\times$ algorithm **interaction was not statistically established**.
4. Divergence analyses are **associational**, not mediation or causal evidence. The regime-interaction survives a granular clustered test, but divergence as a seed-level main effect does not ($p = .446$); the divergence–sensitivity association is $n = 4$.
5. **The original representation $\times$ decision-algorithm matrix does not exceed buy-and-hold** (Sharpe .784) and 2022 is negative for all algorithms. The later exposure-only and hybrid policies improve the tested risk–return trade-off, but **their paired Sharpe advantages are not statistically significant** on the single four-year test period (all paired bootstrap CIs include zero).
6. **Cross-market CSI300 results** were close to unsolvable under the tested protocol, limiting conclusions about representation transfer; its exposure frontier is degenerate (flat dominates).
7. **CQL behaviour depends materially on implementation/regularization choices**; the main-table CQL is the regularized variant.
8. **The regime-gated mechanism is market-specific**: the gate learns a bull-vs-bear asymmetry in every tested market, but the sign of divergence's effect differs (deviation pays in bull on SPY, in crisis on NIFTY), so the "when to deviate" prior does not transfer unchanged.

9. The representation-quality framework evaluates **selected measurable properties** (probe recoverability, rank, CKA, perturbation sensitivity); it is **not a complete measure of the "information contained"** in a representation.
10. **Uncertainty is used in two distinct senses.** Ensemble predictive disagreement did not provide a useful *directional* signal; in contrast, representation-space **OOD distance** emerged as the strongest at-t input in the exposure ablation. These are different constructs and should not be conflated.

7. Conclusion

We separated availability, encoding, and exploitation in offline financial RL and found that, in the tested setting, decision-algorithm identity is the robust determinant of downstream utility, while representation identity is marginal and the representation $\times$ algorithm interaction is not statistically established. Representation diagnostics—probe performance, effective rank, cross-seed stability—do not provide a reliable one-dimensional proxy for trading utility. Behaviour-policy divergence and regime conditioning offer a plausible mechanism for why some algorithms are more representation-sensitive. A pre-specified granular test supports the regime-dependent component of this mechanism (divergence $\times$ regime interaction, $p = 2.7\text{e-}18$) but not divergence as a main effect once algorithm identity is controlled ($p = .446$); the divergence–sensitivity association itself remains descriptive at the algorithm level. Turning from prediction to participation, we then showed that when directional skill is unavailable the exploitable margin is exposure/risk control: an exposure-only policy improved risk-adjusted utility on positive-drift markets, and a continuous risk-aware hybrid recovered most of buy-and-hold's participation while retaining the risk timing (NIFTY Sharpe 1.203 with return .661 vs .694 for buy-and-hold), moving the realized risk–return trade-off above the buy-and-hold ray. This improvement is directionally consistent across markets and seeds and survives cost robustness, but its paired Sharpe advantage is not statistically significant at 5% on the single test period, and it is not an efficient-frontier claim. The strongest final message is: **offline financial RL does not need reliable directional prediction to extract decision-relevant information from learned representations — in the tested setting that information is more useful for adaptive exposure and risk control than for directional trading, while downstream algorithm choice remains a stronger determinant of utility than representation identity.** The findings describe the tested offline environment and do not demonstrate market-beating performance or a causal effect of policy divergence.

Appendix A — CQL implementation and collapse

rep	bc=0 collapse	bc=0 Sharpe	bc=.5 collapse	bc=.5 Sharpe
raw	.60	.568	.00	.574
auto	.30	.256	.00	.768

rep	bc=0 collapse	bc=0 Sharpe	bc=.5 collapse	bc=.5 Sharpe
predictive	.40	.232	.00	.676
contrastive	.50	.291	.00	.709

CQL(H) with a deterministic actor experienced action collapse (action standard deviation $< .01$) in 30–60% of seeds; the TD3+BC actor regularizer eliminated observed collapse and materially improved performance. This demonstrates substantial sensitivity of CQL to actor regularization under our implementation and dataset; it is not a general claim that CQL is unstable.

Appendix B — artifact index

All tables are generated by scripts under `scripts/` and written to `data/interpret/`:

- `rep_scaling.csv`, `rep_scaling_summary.csv` — feature/latent sweeps.
- `rep_offline_rl.csv`, `rep_offline_rl_summary.csv` — representation $\times$ algorithm matrix and baselines.
- `rep_rl_dim_sweep.csv` — algorithm $\times$ latent-dimension.
- `rep_rl_diagnostics.csv`, `rep_divergence_regime.csv` — divergence, action distribution, regime and per-year utility.
- `rep_nonlinear_probe.csv` — linear/GBDT/MLP probes.
- `rep_cross_market.csv` — SPY $\rightarrow$ CSI300 transfer.
- `rep_bootstrap_ci.csv`, `rep_contrasts.csv`, `rep_anova.csv` — statistics.
- `rep_cql_ablation.csv` — CQL regularization ablation.
- `paper/rep_mechanism_schematic.png` — Figure 1.

References

- Agarwal, R., Schwarzer, M., Castro, P. S., & Courville, A. (2021). Deep reinforcement learning at the edge of the statistical precipice. NeurIPS.
- Alain, G., & Bengio, Y. (2017). Understanding intermediate layers using linear classifier probes. ICLR Workshop.
- Chen, L., Lu, K., Rajeswaran, A., Lee, K., Grover, A., Laskin, M., Abbeel, P., Srinivas, A., & Mordatch, I. (2021). Decision Transformer. NeurIPS.
- Chen, T., Kornblith, S., Norouzi, M., & Hinton, G. (2020). A simple framework for contrastive learning of visual representations. ICML.
- Deng, Y., Bao, F., Kong, Y., Ren, Z., & Dai, Q. (2017). Deep direct reinforcement learning for financial signal representation and trading. IEEE TNNLS.
- Fujimoto, S., & Gu, S. S. (2021). A minimalist approach to offline reinforcement learning. NeurIPS.
- Fujimoto, S., Meger, D., & Precup, D. (2019). Off-policy deep reinforcement learning without exploration. ICML.

- Henderson, P., Islam, R., Bachman, P., Pineau, J., Precup, D., & Meger, D. (2018). Deep reinforcement learning that matters. AAAI.
- Janner, M., Li, Q., & Levine, S. (2021). Offline reinforcement learning as one big sequence modeling problem. NeurIPS.
- Jiang, Z., Xu, D., & Liang, J. (2017). A deep reinforcement learning framework for the financial portfolio management problem. arXiv:1706.10059.
- Kornblith, S., Norouzi, M., Lee, H., & Hinton, G. (2019). Similarity of neural network representations revisited. ICML.
- Kostrikov, I., Nair, A., & Levine, S. (2022). Offline reinforcement learning with implicit Q-learning. ICLR.
- Kumar, A., Zhou, A., Tucker, G., & Levine, S. (2020). Conservative Q-learning for offline reinforcement learning. NeurIPS.
- Levine, S., Kumar, A., Tucker, G., & Fu, J. (2020). Offline reinforcement learning: tutorial, review, and perspectives on open problems. arXiv:2005.01643.
- Mnih, V., Badia, A. P., Mirza, M., Graves, A., Lillicrap, T., Harley, T., Silver, D., & Kavukcuoglu, K. (2016). Asynchronous methods for deep reinforcement learning. ICML.
- Moody, J., & Saffell, M. (2001). Learning to trade via direct reinforcement. IEEE Transactions on Neural Networks.
- Pomerleau, D. A. (1991). Efficient training of artificial neural networks for autonomous navigation. Neural Computation.
- Roy, O., & Vetterli, M. (2007). The effective rank: a measure of effective dimensionality. EUSIPCO.
- van den Oord, A., Li, Y., & Vinyals, O. (2018). Representation learning with contrastive predictive coding. arXiv:1807.03748.